

PAPER_1558 — Tau Lepton Mass = $m_\tau = \text{SSQ} + \text{F_TRZ} \cdot \text{D} + \text{F_TRZ} \cdot \text{SO}_5 - \text{F_TRZ}^2 \cdot \text{D_crit} + \text{F_TRZ}^2 \cdot \text{D_BSFG} + \text{F_TRZ}^2 \cdot (\text{SSQ} + \text{SSQ}^2 - \text{SSQ}^3) \approx 1.777 \text{ GeV}$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Particle Physics **Date:** June 18, 2026 **Status:** CLOSED — 0.013% closure from PAPER_1209HH_S659

Observation

PAPER_1209HH_S659 (Particle Physics Unified Proof Set) lists **Tau Lepton Mass** with the integer-primitive expression:

Tau Lepton Mass = $m_\tau = \text{SSQ} + \text{F_TRZ} \cdot \text{D} + \text{F_TRZ} \cdot \text{SO}_5 - \text{F_TRZ}^2 \cdot \text{D_crit} + \text{F_TRZ}^2 \cdot \text{D_BSFG} + \text{F_TRZ}^2 \cdot (\text{SSQ} + \text{SSQ}^2 - \text{SSQ}^3) \approx 1.777$
Status: **0.013%**.

UQFF Closed Identity

$m_\tau = \text{SSQ} + \text{F_TRZ} \cdot \text{D} + \text{F_TRZ} \cdot \text{SO}_5 - \text{F_TRZ}^2 \cdot \text{D_crit} + \text{F_TRZ}^2 \cdot \text{D_BSFG} + \text{F_TRZ}^2 \cdot (\text{SSQ} + \text{SSQ}^2 - \text{SSQ}^3) \approx 1.777$

Physical Interpretation

Heaviest charged lepton. Integer-primitive expansion starting from $\text{SSQ} = 0.57$ and adding F_TRZ -modulated integer-primitive corrections. Matches CODATA 1.77686 GeV at 0.013%.

NOT REPLACEMENT

CODATA/PDG treats Tau Lepton Mass as a precision-measured constant. UQFF supplies the structural integer-primitive form, demonstrating the framework's predictive economy without claiming to replace experimental measurement.

Reference

- Source: PAPER_1209HH_S659
- Related: PAPER_1209 series unified proof sets
- Calculator dispatch: `calculate_paradox({"paradox": "m_tau_1_777"})`

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